

Vaasenk

Mobile App Product Requirements Document

Version 0.1 | Student/Teacher Mobile Experience | Prepared for React Native frontend planning

1. Product Summary

The Vaasenk mobile app is the classroom companion for students and teachers. Students use it to join classrooms, view notes, bookmark/download material, and receive notifications. Teachers use it for quick classroom access, camera-based note uploads, AI assistant access, and lightweight question paper generation/review. Admin-heavy actions such as syllabus repository setup and subscription management belong primarily to the web app.

Decision	Locked specification
Mobile frontend	React Native / Expo + TypeScript recommended
Primary users	Students and Teachers
Admin usage	Not core MVP on mobile; admin may get read-only/limited controls later
Backend	Same Node.js backend used by web app
Database	Same PostgreSQL database
AI access	Teachers can access classroom AI chatbot on mobile after web/admin syllabus setup
Offline	Student note downloads and bookmarks planned for MVP-light or Phase 2 depending timeline

2. Mobile Product Goals

- Make notes accessible instantly after the teacher uploads them.
- Allow teachers to upload board photos from camera in seconds.
- Give students a clean, distraction-free notes feed.
- Support classroom joining through simple invite code.
- Let teachers use the classroom AI assistant from mobile when preparing or answering doubts.
- Reduce dependency on WhatsApp groups for academic material.

Non-goals for Mobile MVP

- Do not build full admin onboarding on mobile.
- Do not build complex analytics dashboards on mobile MVP.
- Do not support student AI chatbot access in MVP.
- Do not build live classes, video calls, or social chat features.
- Do not make mobile UI cluttered with every web feature.

3. User Roles on Mobile

Role	Mobile capabilities
Student	Join class, view classrooms, view notes, bookmark/download notes, ask doubts, receive notifications, manage profile.
Teacher	View assigned classrooms, upload notes using camera/gallery/file, access classroom AI assistant, generate/review question paper, answer doubts.
Institution Admin	Optional later: view institution status, approve users, receive processing alerts. Not MVP focus.

Super Admin	No mobile requirement in MVP.
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4. Mobile Navigation Structure

Role	Navigation
Student	Bottom tabs: Home, Classrooms, Bookmarks, Downloads, Profile. Notification bell in top bar.
Teacher	Bottom tabs: Home, Classrooms, Upload, AI Assistant, Profile. Notification bell in top bar.
Shared	Auth stack: Splash, Welcome, Login, Register/Invite Accept, Forgot Password.

5. Mobile Screen List

Category	Screens
Authentication	Splash, Welcome, Login, Register/Invite Accept, Forgot Password, Role Confirmation.
Student	Student Home, Join Classroom, My Classrooms, Classroom Feed, Note Detail, Bookmarks, Downloads, Ask Doubt, Notifications, Profile, Settings.
Teacher	Teacher Home, My Classrooms, Classroom Detail, Quick Upload, Camera Upload Preview, Upload Note Form, AI Assistant, Generate Paper, Generated Paper Preview, Student Doubts, Notifications, Profile, Settings.
Common states	Loading, Empty, Offline, Error, Permission Denied, File Preview, PDF/Image Viewer.

6. Mobile UI/UX Design System

Element	Specification
Design tone	Clean, fast, friendly, academic, low distraction.
Primary color	Blue #2563EB for trust and education-first feel.
Secondary color	Teal/Green #14B8A6 or #16A34A for completed/success actions.
Background	#F8F AFC light gray for calm reading and card separation.
Text	#0F172A for headings and #475569 for secondary text.
Important tag	#F59E0B amber for exam/important notes.
Danger	#DC2626 only for delete/error actions.
Typography	Inter or Nunito Sans. Use large readable text for student notes.
Cards	Rounded 16px cards, soft shadow, clear subject badges.
Primary CTA	Large thumb-friendly buttons, minimum height 48px.
Accessibility	High contrast, large tap targets, dynamic font support, clear icons with labels.

7. Authentication Requirements

7.1 Mobile Login and Role Routing

Field	Specification
Phase	MVP
Primary users	Teacher, Student
Purpose	Allow teachers and students to securely access the correct mobile experience.

Functional requirements

- User can log in using phone/email and password or institution invite flow.
- System identifies role after login and routes to Teacher Home or Student Home.
- User session persists securely until logout or token expiry.
- If user has multiple roles, app asks which role to continue with.
- Forgot password flow supports OTP/email reset depending implementation.

Acceptance criteria

- Invalid credentials show clear error.
- Student cannot see teacher-only tabs.
- Teacher-only AI and paper generation are hidden for students.
- Expired/disabled user account cannot access app.

Edge cases and error states

- If network is unavailable during login, show offline message.
- If token expires, refresh silently or redirect to login.
- If institution subscription is inactive, show restricted mode message depending role.

UI notes

- Welcome screen should have Vaasenk logo, tagline “Teach more. Copy less.”, and login CTA.

8. Student Mobile Requirements

8.1 Student Home

Field	Specification
Phase	MVP
Primary users	Student
Purpose	Give students quick access to recently uploaded notes and joined classrooms.

Functional requirements

- Student sees greeting, today/recent notes, joined classrooms, unread notifications, and bookmarked notes shortcut.
- Recent notes are sorted by latest upload.
- Student can filter by subject/classroom.
- Student can open any visible note from home.

Acceptance criteria

- Home loads in under 2 seconds under normal network after cached shell is available.
- If no classrooms are joined, show Join Classroom CTA.
- If no notes are available, show friendly empty state.

Edge cases and error states

- If a note is removed, hide from list on refresh.
- If student loses access to classroom, remove notes from home.
- If network is poor, show cached recent content if available.

UI notes

- Use cards: “Continue studying”, “Latest notes”, “My classrooms”. Avoid too many dashboard statistics for students.

8.2 Join Classroom

Field	Specification
Phase	MVP
Primary users	Student
Purpose	Allow students to join a classroom using a code shared by teacher/admin.

Functional requirements

- Student enters classroom invite code.
- System validates code and shows classroom preview before joining.
- Student confirms join.
- Joined classroom appears under My Classrooms.
- If institution uses admin-managed enrollment, code entry may be disabled.

Acceptance criteria

- Invalid code shows error.
- Expired code shows expired message.
- Already joined classroom shows “Already joined”.
- Student cannot join classroom outside allowed institution rules if enforced.

Edge cases and error states

- Code with extra spaces should be trimmed.
- If code belongs to archived classroom, joining is blocked.
- If classroom capacity/user plan limit reached, show contact admin message.

UI notes

- Use simple 6-character code input with paste support and clear CTA.

8.3 Classroom Notes Feed

Field	Specification
Phase	MVP
Primary users	Student
Purpose	Show all notes for a classroom in a clean, searchable, student-friendly feed.

Functional requirements

- Student opens classroom and sees notes grouped by date or chapter/tag.
- Student can filter by Important, Homework, Revision, Exam, Assignment.
- Student can search by note title/description in MVP; OCR full-text search can come in Phase 2.
- Student can bookmark or download note from list.
- Student can open note detail with image/PDF viewer.

Acceptance criteria

- Student sees only notes from enrolled classrooms.
- Bookmark updates immediately.
- Downloaded notes appear in Downloads tab.
- Empty feed shows “No notes uploaded yet.”

Edge cases and error states

- If note preview fails, show file type card.
- If file URL expires, app refreshes signed URL.
- If device storage is full, download fails with clear message.

UI notes

- Feed should prioritize readability: thumbnail, title, subject, teacher, date, tag, bookmark, download.

8.4 Note Detail and Viewer

Field	Specification
Phase	MVP
Primary users	Student, Teacher
Purpose	Let students read classroom material comfortably on mobile.

Functional requirements

- Support image viewer with pinch zoom.
- Support PDF viewer or open-in-browser fallback.
- Show title, description, tags, teacher name, upload date.
- Allow bookmark, download, share if enabled, and ask doubt if enabled.
- For text notes, show clean reading layout.

Acceptance criteria

- Viewer opens from feed without losing classroom context.
- Large images remain zoomable and readable.
- Downloaded note can be opened offline if already stored locally.
- If note access is revoked, viewer blocks content.

Edge cases and error states

- If PDF is too large, show download/open externally option.
- If offline and note not downloaded, show “Connect to internet to view.”
- If file is corrupted, show unavailable message.

UI notes

- Use full-screen viewer with top back button and bottom action bar.

8.5 Bookmarks and Downloads

Field	Specification
Phase	MVP
Primary users	Student
Purpose	Help students revise important notes quickly and access material with poor internet.

Functional requirements

- Student can bookmark notes from feed/detail.
- Student can download notes if allowed by teacher/institution.
- Downloads tab lists downloaded files by classroom and subject.
- App indicates whether a note is available offline.
- User can remove downloaded file from device.

Acceptance criteria

- Bookmarks sync across devices through backend.
- Downloaded file stays on device until removed or app cache cleared.
- Download progress is visible.
- Failed download can be retried.

Edge cases and error states

- If storage permission is denied, use app-private storage where possible or show permission instruction.
- If institution disables downloads, hide download button.

- If file is updated, app should indicate newer version available.

UI notes

- Use clear icons: bookmark, download, offline badge.

8.6 Ask Doubt Under Note

Field	Specification
Phase	MVP
Primary users	Student, Teacher
Purpose	Allow students to ask questions without disturbing the live class.

Functional requirements

- Student can submit a doubt under a specific note.
- Teacher can view and reply to doubts.
- Student receives notification when teacher replies.
- Doubt can be marked Resolved by teacher or student.
- MVP can support text-only doubts.

Acceptance criteria

- Doubt is linked to note and classroom.
- Only classroom teacher/admin can view all doubts.
- Student can view own doubts and replies.
- Empty doubts section invites student to ask a question.

Edge cases and error states

- If note is deleted, retain doubt history but mark note unavailable.
- If student is removed from classroom, future access to doubts is blocked.
- Abusive content moderation can be Phase 2.

UI notes

- Keep it simple: question box, send button, reply thread. No public social comments in MVP.

9. Teacher Mobile Requirements

9.1 Teacher Home

Field	Specification
Phase	MVP
Primary users	Teacher
Purpose	Give teachers immediate access to assigned classrooms, pending doubts, and quick upload.

Functional requirements

- Teacher sees assigned classrooms, recent uploads, pending student doubts, and AI processing alerts.
- Teacher has visible Quick Upload CTA.
- Teacher can open classroom detail from home.
- Teacher can access AI Assistant shortcut for recent classroom.

Acceptance criteria

- Teacher sees only assigned classrooms.
- Pending doubts count is accurate.
- Quick Upload requires classroom selection if launched from home.

Edge cases and error states

- If teacher has no classrooms, show “No classrooms assigned yet. Contact admin.”
- If syllabus is not AI-ready, AI shortcut shows disabled status.

UI notes

- Teacher home should be action-first: Upload Note, Open Classroom, AI Assistant, Pending Doubts.

9.2 Quick Camera Note Upload

Field	Specification
Phase	MVP
Primary users	Teacher
Purpose	Let teachers upload board photos immediately from mobile.

Functional requirements

- Teacher can select classroom and capture photo using camera.
- Teacher can upload from gallery or file picker.
- Teacher can crop/rotate image before upload in Phase 2; MVP can use preview only.
- Teacher adds optional title, description, and tags.
- Teacher publishes note or saves as draft.
- Students receive notification after publish.

Acceptance criteria

- Camera permission prompt appears only when needed.
- Image preview appears before upload.
- Upload progress is visible.
- Failed upload can be retried.
- Published note appears in classroom feed.

Edge cases and error states

- If permission denied, show instructions to enable camera access.
- If image is too large, compress before upload or show size error.
- If teacher leaves app during upload, upload should continue if feasible or resume/retry gracefully.

UI notes

- This should be the smoothest teacher flow: Open app > Upload > Camera > Preview > Publish.

9.3 Teacher Classroom Detail

Field	Specification
Phase	MVP
Primary users	Teacher
Purpose	Provide mobile access to classroom content and teacher actions.

Functional requirements

- Teacher can view classroom notes, students, doubts, generated papers, and AI assistant.
- Teacher can upload new note from classroom detail.
- Teacher can reply to student doubts.
- Teacher can open generated question papers.
- Teacher can see syllabus and AI status but cannot manage syllabus from mobile MVP.

Acceptance criteria

- Classroom detail clearly shows subject, class, section, student count, and syllabus status.
- Teacher-only actions are visible to teacher and hidden from students.
- If AI is not ready, show clear reason and ask admin to fix syllabus if needed.

Edge cases and error states

- If classroom is archived, upload and AI generation are disabled.
- If teacher removed from classroom, detail access is revoked on refresh.

UI notes

- Use top segmented tabs: Notes, Doubts, Papers, AI.

10. Mobile AI Assistant Requirements

10.1 Teacher AI Assistant on Mobile

Field	Specification
Phase	MVP
Primary users	Teacher only in MVP
Purpose	Allow teachers to chat with classroom-specific syllabus AI from mobile.

Functional requirements

- Teacher opens AI tab inside classroom or global AI tab then selects classroom.
- Chatbot uses mapped syllabus and sample paper guidance prepared from web/admin setup.
- Teacher can ask for chapter summaries, lesson plans, important questions, simplified explanations, quick quizzes, and doubt explanations.
- Teacher can copy response or save as note draft in Phase 2.
- Chat history is saved by classroom and teacher.
- Quick prompt chips appear above keyboard: Summary, Important Questions, Lesson Plan, Quiz, Explain Simply.

Acceptance criteria

- AI tab is hidden from students.
- Teacher cannot access AI for classrooms not assigned to them.
- If syllabus is Processing/Failed/Not Mapped, chat input is disabled with explanation.
- Response should include source references where available.
- Teacher can report wrong answer.

Edge cases and error states

- If AI service timeout occurs, show retry.
- If answer is outside syllabus context, assistant must clearly say so.
- If teacher loses network, queued message should not duplicate after retry.
- If AI credits are exhausted, show institution/admin contact message.

UI notes

- Use familiar chat UI but keep it academic: no public chat feel, no student-like reactions, no social clutter.

11. Mobile Question Paper Flow

11.1 Generate or Review Question Paper on Mobile

Field	Specification
Phase	MVP
Primary users	Teacher
Purpose	Give teachers a lightweight way to generate or review question papers from mobile.

Functional requirements

- Teacher can open Generate Paper from classroom.
- System uses classroom mapped syllabus automatically.
- Teacher selects portion, exam type, total marks, question types, difficulty, and answer key option.
- Teacher can use default blueprint templates created on web/admin side.
- Generated output can be previewed on mobile.
- Teacher can save draft and download/share PDF.

Acceptance criteria

- Mobile flow should use default templates to reduce form burden.
- Generate disabled if syllabus is not AI Ready.
- Teacher can view status of generation job.
- Generated paper is accessible from Papers tab.

Edge cases and error states

- If long generation takes time, app may send notification when ready.
- If form is too complex, suggest using web app for advanced configuration.
- If selected marks pattern is invalid, show correction prompt.

UI notes

- Mobile should focus on simple presets. Advanced blueprint editing belongs to web.

12. Notifications on Mobile

12.1 Push and In-App Notifications

Field	Specification
Phase	MVP
Primary users	Student, Teacher
Purpose	Notify students and teachers about classroom activity without WhatsApp dependency.

Functional requirements

- Student receives notification for new note, new published paper, teacher doubt reply, classroom announcement.
- Teacher receives notification for student doubt, syllabus AI-ready status, question paper generation status, upload failure.
- Notification opens related screen.
- User can mark notification as read.
- User can configure notification preferences.

Acceptance criteria

- Unread count appears on bell icon.
- Push notification is optional if device permission granted.
- In-app notifications work even if push permission denied.

Edge cases and error states

- If notification target no longer exists, open fallback message.
- If user disabled notifications, show only in-app list.
- If classroom access removed, notification should not reveal private content.

UI notes

- Use short, clear notification copy. Example: "New Maths note uploaded in 10-A."

13. Offline and Sync Requirements

- MVP-light: bookmarks sync online; downloaded files are available offline.
- Phase 2: cached classroom feed and note metadata available offline.
- Do not allow offline note upload in MVP unless upload queue is intentionally built.
- If user opens app offline, show Offline Mode banner.
- Offline downloaded notes must still respect logout/security rules where possible.
- When internet returns, refresh signed URLs, notifications, and bookmark status.

14. Mobile API Usage Rules

- Mobile app must call only Vaasenk Node.js backend APIs.
- Mobile app must never call AI provider, Flowise, or Langflow directly.
- Use short-lived access token and refresh token flow.
- Use pagination for notes feed and notifications.
- Use signed URLs for file upload/download.
- Use background-safe upload strategy for note photos.
- All AI requests must include classroom ID and pass backend permission checks.

15. Mobile State and Empty State Requirements

State	Required message/action
No classrooms	"You have not joined any classroom yet." CTA: Join Classroom.
No notes	"No notes uploaded yet. Check again after class."
No bookmarks	"Bookmark important notes to find them quickly."
No downloads	"Download notes to access them offline."
AI not ready	"Syllabus is still being prepared for AI. Try again later."
AI failed	"AI setup failed. Ask your admin to reprocess syllabus."
No camera permission	"Camera permission is required to upload board photos." CTA: Open Settings.
Offline	"You are offline. Downloaded notes are still available."
Upload failed	"Upload failed. Check connection and retry."

16. Performance Requirements

- Initial app shell should feel fast on budget Android phones.
- Use image compression for teacher camera uploads.
- Use lazy loading for classroom feeds and thumbnails.
- Use pagination/infinite scroll for notes and notifications.
- Cache recent classroom list and downloaded files.
- Avoid heavy animations that affect low-end devices.
- File upload should show progress and not freeze UI.
- Chat response should support streaming later, but MVP can show loading and full response.

17. Security Requirements for Mobile

- Store tokens securely using secure storage, not plain AsyncStorage.
- Lock content access through backend permissions, not only frontend hiding.
- Disable screenshots only if institution demands it later; not MVP default.
- Do not store sensitive AI keys in mobile app.
- Downloaded files should use app-private storage when possible.
- On logout, clear cached private metadata and optionally downloaded files based on settings.
- Do not show classroom content in push notification body if privacy mode is enabled.

18. Mobile Development Phases

Phase	Features
Phase 1: UI MVP	Auth, role routing, student home, teacher home, classrooms, notes feed, note detail, upload flow, AI assistant screen, generate paper screen, notifications, profile/settings.
Phase 2: Backend integration	Login, classroom list, join code, notes APIs, upload signed URLs, bookmarks, notifications, teacher doubts.
Phase 3: AI integration	Teacher AI chat, generate paper jobs, AI status badges, push completion notifications.
Phase 4: Offline	Downloads, cached metadata, offline viewer, sync improvements.
Phase 5: Advanced	Student AI assistant, flashcards, OCR search, parent login, analytics snippets.

19. Final Mobile MVP Lock

- Students use mobile app mainly for classroom notes, bookmarks, downloads, doubts, and notifications.
- Teachers use mobile app for quick note/photo upload and classroom AI access.
- Admin syllabus upload and sample paper setup stay on web app.
- Teacher AI assistant is available on mobile only after admin/web setup is complete.
- Question paper generation on mobile should be template-based and simpler than web.
- Mobile app must remain fast, clean, and focused. It should not become a full ERP dashboard.